

Lesson 20 Functions

We are familiar with a function as an input-output system with a rule or formula. The function $f(x) = x^2$ takes in a number x and outputs the number x^2 , $f(2) = 2^2 = 4$. The function $g(x, y) = x + y$ takes in 2 numbers and give the sum: $g(1, 3) = 1 + 3 = 4$. In each case, an input is **mapped** to (sent to) an output:

$$2 \longrightarrow 4, \quad (1, 3) \longrightarrow 4$$

We will see that in a formal study a function is more than a rule or formula.

Definition A **function** f from a set A to a set B ($f: A \rightarrow B$), is a rule which assigns each $x \in A$ to *exactly one element* $f(x)$ in B ; A and B are the **domain** and **codomain** of f . The **image** of f is a subset of B : $\text{im } f = \{f(x) : x \in A\}$.

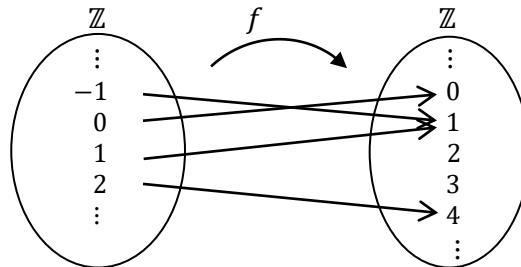
There are 3 components in a complete function definition: A function $f: A \rightarrow B$ is defined only if its domain A and codomain B are specified along with a good rule f . A good rule f assigns each $x \in A$ to *exactly one element* $f(x)$ in B :

- If $x \in A$, then $f(x) \in B$.
- If $x, y \in A$ and $x = y$, then $f(x) = f(y)$. This means that f is **well-defined**.

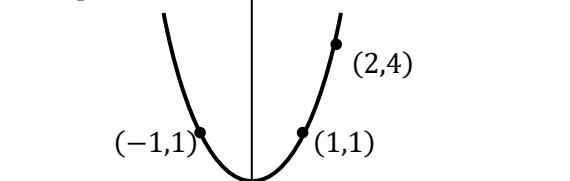
Two functions are **equal** if they have the same domain, codomain, and rule.

- a) The function $f: \mathbb{Z} \rightarrow \mathbb{Z}$, $f(x) = x^2$, is good: it maps an integer x to exactly one integer x^2 . The image is the set of all “squares”: $\text{im } f = \{x^2 : x \in \mathbb{Z}\} \subseteq \mathbb{Z}$.

We may use ovals for A and B , and arrows to show where an element is sent.



- b) The function $f: \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = x^2$, is good: each real number is sent to exactly one number. The image is the nonnegative numbers: $\text{im } f = \mathbb{R}_{\geq 0} = \{x \in \mathbb{R} : x \geq 0\}$. A function with $\mathbb{R}$ as the domain and codomain is a **real function**, whose **graph** is the set of points $\{(x, x^2) : x \in \mathbb{R}\}$ in the Cartesian plane $\mathbb{R} \times \mathbb{R}$.



- c) $f: \mathbb{Z} \rightarrow \mathbb{N}$, $f(x) = x^2$, is *not* a valid function: $0 \in \mathbb{Z}$ but $f(0) = 0 \notin \mathbb{N}$; not every element in the domain is mapped to an element in the codomain.
- d) For a *well-defined* square-root function, restrict $f(x) = \sqrt{x}$ to be *nonnegative*.

A function $f: A \rightarrow B$ defines a set of *ordered pairs* $(a, f(a))$, we may say that $f = \{(a, f(a)): a \in A\} \subseteq A \times B$, then $f: A \rightarrow B$ is a relation from A to B .

Definition 2 A **function** f from set A to set B is a relation $f \subseteq A \times B$ such that:

- for every $x \in A$, $(x, y) \in f$ for some $y \in B$
- if $(x, y) \in f$ and $(x, z) \in f$, then $y = z$

The two definitions of function coincide exactly with the equivalent notations:

$$(x, y) \in f \quad \leftrightarrow \quad y = f(x)$$

In the first definition we have a rule f . In the second definition, the rule is the set of ordered pairs f . For example, there are two ways to define the square function rule on $\mathbb{Z}$: $f(x) = x^2$, or $f = \{..., (-2, 4), (-1, 1), (0, 0), (1, 1), (2, 4), ...\}$.

Every function is a relation but not every relation is a function.

The set $R = \{(1, 3), (2, 3)\}$ is a relation from $A = \{1, 2, 3\}$ to $B = \{3\}$ but R is *not* a function from A to B , because $3 \in A$ but $(3, y) \notin R$ (for any y): 3 is not mapped to any element in B ! The set $R = \{(1, 2), (1, 3)\}$ can be a relation from a set which contains 1 to a set which has 2 and 3, but it cannot be a function since $(1, 2) \in R$ and $(1, 3) \in R$ but $2 \neq 3$, *not well-defined*.

Recall that the **image of a function** is the subset of codomain which is “hit”. We now introduce the **image of a subset** (of the domain) under a function.

Definition Let A and B be sets and $f: A \rightarrow B$. Let $S \subseteq A$ and $T \subseteq B$.

- The **image** of S under f is $f(S) = \{f(x) : x \in S\}$.
- The **inverse image** of T under f is $f^{-1}(T) = \{x \in A : f(x) \in T\}$.

If $S = A$, then $f(S) = f(A) = \text{im } f$ and we use $\text{im } f$, as the emphasis is on f , rather than S .

Let $A = \{1, 2, 3\}$, $B = \{1, 2, 3, 4, 5\}$, and $f: A \rightarrow B$ be defined by $f(x) = x + 1$. The image of f is $\{2, 3, 4\}$. Let's take a subset of the domain: $S = \{1, 2\}$, then the **image of S under f** is $\{2, 3\}$, denoted $f(S) = \{2, 3\}$, and the inverse image of $\{2, 3\}$ is $f^{-1}(\{2, 3\}) = \{1, 2\}$.

Proposition Let $f: A \rightarrow B$ and $R, S \subseteq A$, then $f(R \cap S) \subseteq f(R) \cap f(S)$.

Proof Let $y \in f(R \cap S)$, then $y = f(x)$ for some $x \in R \cap S$.

Therefore $y = f(x)$ for some $x \in R$, and $y = f(x)$ for some $x \in S$.

So $y \in f(R)$ and $y \in f(S)$, that is, $y \in f(R) \cap f(S)$.

However, it is not true that $f(R \cap S) = f(R) \cap f(S)$, we disprove with a counterexample. Let $A = \{1, 2\}$, $f(1) = f(2) = 1$, and $R = \{1\}$ and $S = \{2\}$. Then $R \cap S = \emptyset$, $f(R \cap S) = \emptyset$, but $f(R) \cap f(S) = \{1\} \cap \{1\} = \{1\}$.